Asteria Space Station: Occupational Safety Analytics



Analyzing injury trends, severity, and prevention priorities, 2083-2087

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Synthetic workplace safety data

1. Project overview

Asteria Space Station is a fictional commercial orbital facility whose workforce supports engineering and maintenance, cargo and logistics, research, habitat services, medical services, and extravehicular operations.

This project uses 1,250 synthetic injury cases covering 2083-2087, together with workforce exposure data, to help station leadership prioritize injury prevention. The Power BI analysis compares injury frequency, exposure-adjusted rates, and injury outcomes across eight operational divisions, then examines EVA (extravehicular activity) & Exterior Operations in greater detail.

The central question is: Where should station leadership prioritize injury prevention, based on injury frequency, severity, and patterns across divisions during 2083-2087?

All organizational, incident, and exposure data are fictional and were generated with ChatGPT for this portfolio project.

2. Data and approach

The analysis uses an incident table containing one row per injury case, an exposure table containing workforce and hours-worked information by division, and supporting date, year, division, location, and occupation tables. Power BI relationships connect these tables to support comparisons across organizational groups and time periods.

Data Quality and Model Validation

In Power Query, I reviewed all 1,250 incident IDs and confirmed that none were missing or duplicated. I also checked incident dates and days-away and restricted/transferred-day fields for missing values, errors, and invalid ranges. These checks identified no issues requiring correction in the fields reviewed.

During analysis, I identified and resolved two filtering issues. Using Year from Dim_Date initially filtered incidents without filtering exposure hours, understating annual rates. Replacing it with Year from Dim_Year aligned annual case counts and hours worked. An inactive occupation-to-incident relationship also caused totals to repeat across occupations; activating the direct Occupation ID

relationship and deactivating the division-to-occupation relationship enabled occupation-level comparisons.

Measures and Analytical Approach

I compared total injuries, recordable cases, days away from work, and two exposure-adjusted rates:

Total Recordable Incident Rate (TRIR) = (OSHA-recordable cases X 200,000) / Total hours worked

Days Away, Restricted, or Transferred (DART) Rate = (Recordable cases involving days away, restricted duty, or job transfer X 200,000) / Total hours worked

Each qualifying case is counted once in DART. Both calculations use cases and hours from the same division and reporting period. The report's "Total Lost Workdays" measure represents days away from work; restricted/transferred days are separate.

The analysis progressed from station-wide division comparisons to EVA's annual trends, event categories, occupations, and individual case descriptions. Fatalities were considered separately because days-away totals alone do not capture the full severity of outcomes. Corrective-action descriptions and statuses informed recommendations, while their effectiveness remains unverified.

3. Findings

EVA warranted closer investigation

Engineering & Maintenance recorded the most injuries, with 203 cases. However, EVA & Exterior Operations had the highest five-year TRIR (8.65) and DART rate (2.01), compared with station-wide rates of 3.69 and 0.71. These exposure-adjusted results identified EVA as a priority for further investigation.

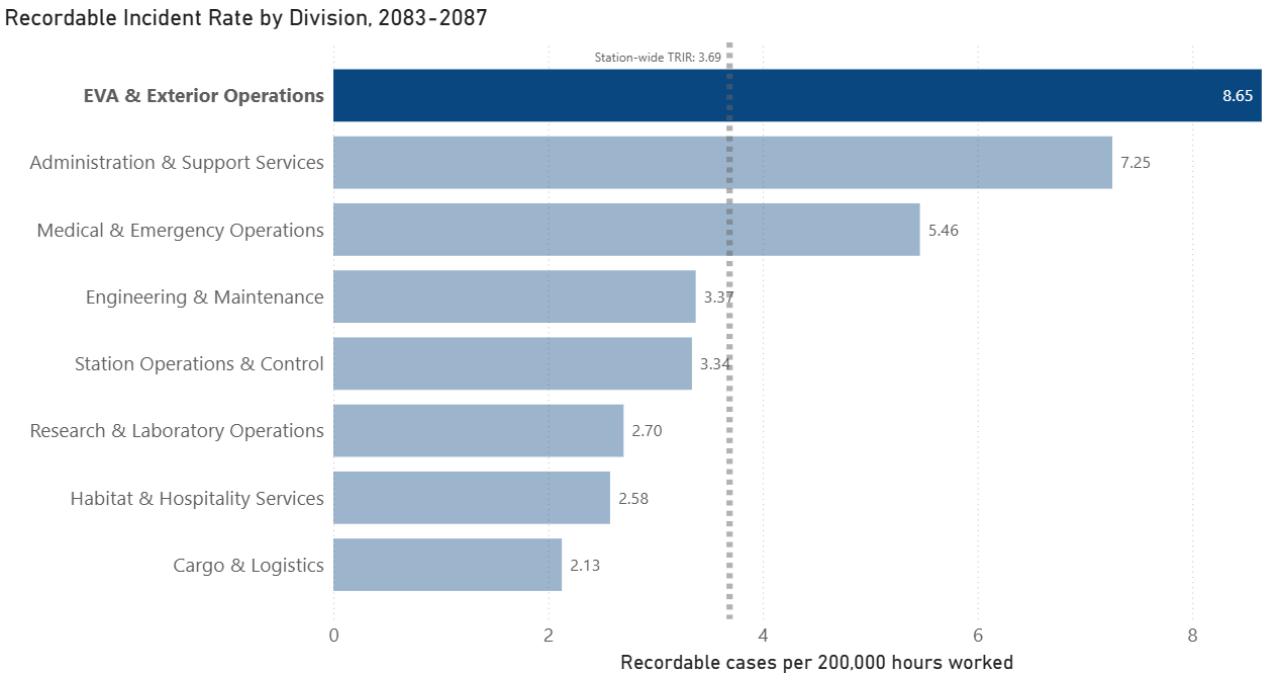


Figure 1. Five-year recordable incident rates by division, 2083-2087. EVA & Exterior Operations had the highest rate of 8.65, compared with the station-wide rate of 3.69. Source: synthetic Asteria data.

Annual performance improved, but concerns remained

EVA's TRIR declined from 12.35 in 2086 to 6.76 in 2087, while its DART rate fell from 2.90 to 0.68. Days away from work decreased from 39 to 13 despite an increase in hours worked. These results indicate improved recorded performance, but do not establish its cause or whether the improvement would continue.

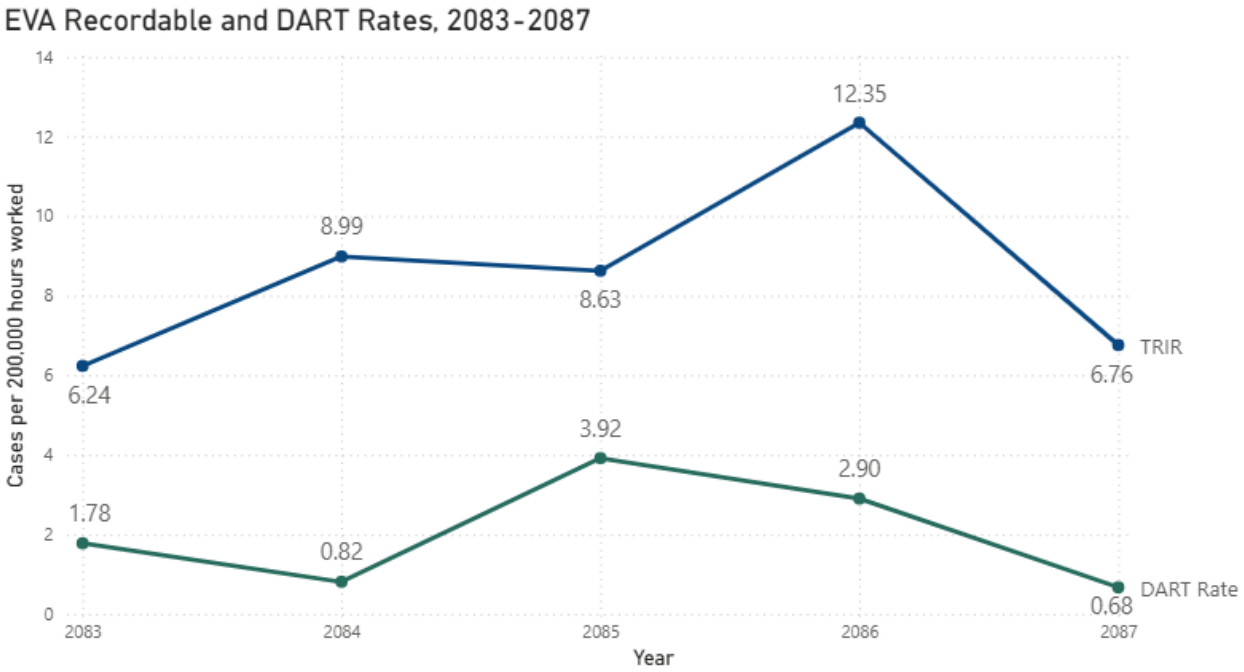


Figure 2. Annual TRIR and DART rates for EVA & Exterior Operations, 2083-2087. Both rates declined in 2087 following peaks in different years. Rates represent cases per 200,000 hours worked. Source: synthetic Asteria data.

Overexertion accounted for most days away

Overexertion accounted for 45 of EVA's 157 injuries (29%), 20 of its 56 recordable cases (36%) and 50 of its 92 days away from work (54%).

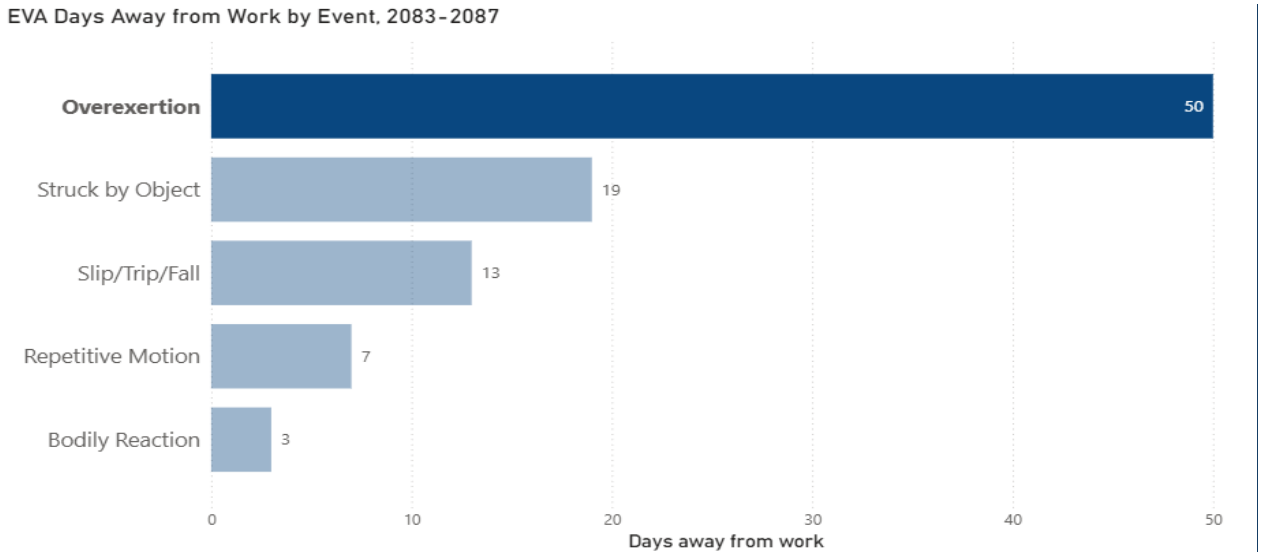


Figure 3. Days away from work by event category in EVA & Exterior Operations, 2083-2087. Overexertion accounted for 50 of 92 days away (54%). Days away do not capture all severe outcomes, including fatalities. Categories with zero days away are omitted. Source: synthetic Asteria data.

All 50 days were associated with three exterior access equipment inspection cases involving EVA Equipment Technicians and EVA Operations Coordinators. The cases occurred in 2085, 2086, and 2087, resulting in 23, 14, and 13 days away, respectively. The 2087 case accounted for EVA's entire annual days-away total.

Each case recorded an action to review handling methods and evaluate mechanical assistance. One action was completed and two were in progress. The records did not establish what changes were implemented or whether they were effective.

Inspection-related overexertion cases accounting for 50 days away, 2085-2087				
Incident ID	Year	Body part	Days away	Corrective-action status
INC-00045	2085	Knee	23	In Progress
INC-00366	2086	Back	14	Completed
INC-00239	2087	Hand	13	In Progress

Table 1. Source: synthetic Asteria data.

A fatal maintenance event revealed a separate critical concern

In 2086, a technician sustained fatal injuries after being struck by a component released under pressure during maintenance on an atmospheric pressure regulation system.

Recorded corrective actions included redesigned pressure-relief safeguards, redundant pressure-monitoring sensors, revised hazardous-energy isolation procedures, and verification of zero-energy conditions before maintenance. The corrective-action status was Open.

This fatality demonstrates why injury rates and days-away totals must be considered alongside individual severe outcomes when setting prevention priorities.

4. Recommendations and limitations:

Priority 1: Follow up on the fatality-related corrective action

Prioritize the open corrective action associated with the 2086 fatal maintenance incident. Request evidence that the recorded pressure-control safeguards and revised hazardous-energy isolation procedures were implemented and verified. The description of planned or recorded actions alone does not establish completion or effectiveness.

Priority 2: Assess exterior access equipment inspection tasks

Investigate the physical demands of the inspection task associated with three overexertion cases and 50 days away from work. Follow up on the two corrective actions in progress and review evidence supporting the completed action. Evaluate whether changes to handling methods or mechanical assistance address the task's identified exposures.

Monitor results

Track recordable cases, DART cases, days away, and severe outcomes alongside corrective-action progress. Review these measures together so that improvements in overall rates do not obscure recurring tasks or potentially fatal exposures.

Limitations

All data are synthetic, and the findings do not represent an actual organization. The detailed investigation focused on EVA & Exterior Operations; it does not constitute a comprehensive assessment of every division.

The three inspection-related cases identify a recurring task associated with injury impact, but their descriptions do not establish a shared root cause. Occupation-specific exposure hours were not available in the reviewed exposure table, so occupation counts cannot establish relative injury risk.

Corrective-action implementation dates, due dates, and effectiveness evidence were not established. Open actions therefore cannot be classified as overdue, and the improvement observed in 2087 cannot be attributed to the recorded actions. The detailed impact analysis emphasized days away rather than a complete assessment of restricted/transferred days.